

Substrate Hall-algebra engine v0.2 — the canonical/crystal basis of $U_q^+(B_2)$: deriving the elementary-particle basis (the assigned→derived lever)

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0. Why this is the lever

Every Periodic-Table cell is currently flagged “assigned” — we hand-picked the particle 5-tuples ($\text{Region} \times \sigma_{\text{BF}} \times \text{Chirality} \times \text{Charge} \times \text{Winding}$). The canonical/crystal basis of $U_q^+(B_2)$ (Lusztig-Kashiwara) is the mathematics that derives the natural basis of the algebra. If the elementary particles ARE the canonical basis elements, the taxonomy stops being a labeling and becomes a theorem. This is engine step 2/4 (after the coproduct, v0.1).

1. The canonical basis structure (rigorous, standard Lusztig-Kashiwara)

1.1 Convex order from w_0

$w_0(B_2) = s_1 s_2 s_1 s_2$ (length 4 = $|\Phi^+|$). It induces the convex order of positive roots:

$\beta_1 = \alpha_1$ (short, ht 1) $\preceq \beta_2 = 2\alpha_1 + \alpha_2$ (long, ht 3, = highest root θ) $\preceq \beta_3 = \alpha_1 + \alpha_2$ (short, ht 2) $\preceq \beta_4 = \alpha_2$ (long, ht 1)

1.2 PBW / canonical basis

Relative to this convex order, the PBW basis of $U_q^+(B_2)$ is

$$E_{\{\beta_1\}}^{c_1} E_{\{\beta_2\}}^{c_2} E_{\{\beta_3\}}^{c_3} E_{\{\beta_4\}}^{c_4}, (c_1, c_2, c_3, c_4) \in \mathbb{Z}_{\geq 0}^4$$

and Lusztig’s canonical basis B agrees with this PBW basis up to lower terms in the order. So $B(\infty) \cong \mathbb{Z}_{\geq 0}^4$ — the canonical basis is parametrized by Lusztig data (c_1, c_2, c_3, c_4) (one exponent per positive root).

1.3 The two defining properties (why “canonical”)

- Bar-invariance: the canonical basis is fixed by the bar involution ($q \leftrightarrow q^{-1}$) — it’s the self-dual integral basis.
- Positivity (Lusztig): the structure constants of the canonical basis are in $\mathbb{Z}_{\geq 0}[q, q^{-1}]$ — non-negative integer (Laurent) polynomials. This positivity is what makes the structure constants count something — they are honest multiplicities, which is precisely the property a physical vertex amplitude needs.

2. What it derives (the particle basis)

2.1 The 4 elementary excitations (finite B_2)

The canonical basis elements at unit Lusztig data (the 4 root vectors) are the elementary excitations: $-E_{\{\alpha_1\}}$ — short simple root vector $-E_{\{\alpha_2\}}$ — long simple root vector $-E_{\{\alpha_1 + \alpha_2\}}$ — short non-simple $-E_{\{2\alpha_1 + \alpha_2\}}$

— long non-simple (highest root)

These are the 4 indecomposables (= positive roots) — the elementary-particle list of the finite algebra, derived as the degree-(positive-root) canonical basis elements, not assigned.

2.2 The crystal graph = the substrate's state ladder

The Kashiwara operators $\tilde{f}_1, \tilde{f}_2$ act on the crystal $B(\infty)$, generating the crystal graph (a colored directed graph on the canonical basis). Physically: - $\tilde{f}_i$ = the substrate's lowering operator (add a quantum of root α_i) — the emission/absorption ladder between states. - The crystal graph IS the substrate's transition network among elementary + composite states, at the combinatorial ($q \rightarrow 0$) level.

This is the canonical-basis refinement of the step-1 coproduct: the coproduct gave decay vertices; the crystal operators give the ladder structure that organizes them.

2.3 Composite / multi-particle states

Canonical basis elements with $|c| > 1$ (higher Lusztig data) = composite/multi-particle states, built canonically (with positive integer multiplicities) from the elementary ones. The canonical basis automatically organizes the multi-particle tower — and (per step 1) the coproduct resolves each into constituents.

3. The affine tower (the generation direction)

For the full substrate we need affine $\hat{B}_2$ (per #407 / E1 — finite B_2 's 4 indecomposables are too few). The canonical basis of $U_q^+(\hat{B}_2)$ has: - a REAL-root part (preprojective/preinjective — the “discrete” excitations), and - an IMAGINARY-root part (the δ -tower — the regular/tube modules).

The imaginary-root (δ -direction) part of the canonical basis is the generation tower (Elie's imaginary-root $\leftrightarrow$ mass-spectrum lane), and the exceptional tubes (#407) sit here. So the canonical basis is exactly where the generation structure becomes a derived basis — the assigned $\rightarrow$ derived conversion for the *winding-mode* coordinate happens here.

4. The assigned $\rightarrow$ derived conversion (what flips, what's still flagged)

5-tuple coordinate	Canonical-basis derivation status
(which elementary excitation)	DERIVED — the 4 root-vector canonical basis elements (finite) + affine tower
Winding / generation	DERIVED-pending — the δ -tower / exceptional tubes (needs #407 pin + Elie imaginary-root)
Region (Shilov/Bulk)	NOT yet — needs the $A_{\text{sub}} \leftrightarrow H(Q_{B2})$ dictionary (Phase 2): which canonical elements are boundary vs bulk
σ_{BF} (bose/fermi)	NOT yet — the Z_2 grading on the canonical basis (Phase 2)
Chirality (L/R)	NOT yet — Spin(5)-cover refinement (Phase 2)
Charge	NOT yet — Gell-Mann-Nishijima from the grading + σ_{BF} (Phase 2)

Honest state: the canonical basis DERIVES the *elementary excitation list* + the *ladder* + (pending #407) the *generation tower*. The PHYSICAL 5-tuple coordinates (region/ σ_{BF} /chirality/charge) still require the $A_{\text{sub}} \leftrightarrow H(Q_{B2})$ dictionary (Phase 2) — that dictionary is what maps the algebra's canonical basis onto the physical coordinate axes. So this flips the “which particle” and “generation” cells toward derived; the region/charge/spin cells await the dictionary.

5. Connection to the program

- Goal 1 (full Hall algebra): the canonical basis is the natural basis in which the full structure-constant table (Goal 1) has positive integer entries — the “right” coordinates for the engine.
- Goal 2 (process model): positivity = the vertex multiplicities are honest counts; the crystal graph = the transition ladder.
- Goal 3 (Periodic Table): this is the lever that flips cells from assigned to derived — the elementary list + generation tower become derived; the table’s “derived” column grows.
- Feeds: Elie’s reaction table (#405 — canonical basis = the right basis for it) + imaginary-root tower; Grace’s Periodic Table derived-flags; the #407 tube structure (the δ -part of the canonical basis).

6. Honest scope + tier

RIGOROUS (standard Lusztig-Kashiwara): w_0 convex order; PBW = canonical up to lower terms; $B(\infty) \cong Z^4_{\geq 0}$ Lusztig data; bar-invariance; positivity ($Z_{\geq 0}[q, q^{-1}]$ structure constants); crystal operators $\tilde{f}_i$; the real/imaginary split for the affine case.

DERIVED (this doc’s claim): the elementary-excitation list = the 4 root-vector canonical basis elements (finite) — not assigned. The ladder = crystal graph.

FRAMEWORK / PENDING: - The affine δ -tower = generation tower (rides on #407 Dlab-Ringel pin + Elie imaginary-root). - The physical 5-tuple mapping (region/ σ_{BF} /chirality/charge) — needs the $A_{sub} \leftrightarrow H(Q_{B2})$ dictionary (Phase 2). This is the boundary between “algebra basis derived” and “physical labels derived.”

Cal #27 / honesty: I am NOT claiming the canonical basis derives the full 5-tuple — it derives the *algebra’s* natural basis (elementary list + ladder + generation-tower-pending), which is the lever; the physical-coordinate mapping is the separate Phase-2 dictionary. The positivity is the deep reason these are physical (they count), but “counts $\mathbb{Z}$ is a particle” is the bet, earned via the dictionary.

Next (my lane): engine step 3 — the R-matrix (the scattering S-matrix from the quasi-triangular structure / the braiding on the canonical basis). Then step 4 — the negative part (antiparticles, Drinfeld double). And the $A_{sub} \leftrightarrow H(Q_{B2})$ dictionary (Phase 2) is the gate that converts the remaining 5-tuple cells to derived.

— Lyra, Engine v0.2 canonical/crystal basis filed (Phase I step 2/4). The assigned→derived LEVER: the canonical basis of $U_q^+(B_2)$ DERIVES the elementary-excitation list (4 root vectors, finite; + affine δ -tower for generations) with Lusztig POSITIVITY (structure constants count — the property a physical vertex needs), parametrized by Lusztig data $Z^4_{\geq 0}$; crystal operators = the state ladder. Flips the “which particle” + “generation” Periodic-Table cells toward derived. Physical 5-tuple coordinates (region/ σ_{BF} /chirality/charge) still need the $A_{sub} \leftrightarrow H(Q_{B2})$ dictionary (Phase 2). Next: R-matrix (step 3).